

randomness: the reason my cup of coffee does not jump is that the sum of all of its moving particles becomes smooth. Likewise, you reach certainties by adding up small Gaussian uncertainties: this is the law of large numbers.

The Gaussian is not self-similar, and that is why my coffee cup does not jump on my desk.

Now, consider a trip up a mountain. No matter how high you go on the surface of the earth, it will remain jagged. This is even true at a height of 30,000 feet. When you are flying above the Alps, you will still see jagged mountains in place of small stones. So some surfaces are not from Mediocristan, and changing the resolution does not make them much smoother. (Note that this effect only disappears when you go up to more extreme heights. Our planet looks smooth to an observer from space, but this is because it is too small. If it were a bigger planet, then it would have mountains that would dwarf the Himalayas, and it would require observation from a greater distance for it to look smooth. Likewise, if the planet had a larger population, even maintaining the same average wealth, we would be likely to find someone whose net worth would vastly surpass that of Bill Gates.)

[Figures 11](#) and [12](#) illustrate the above point: an observer looking at the first picture might think that a lens cap has fallen on the ground.

Recall our brief discussion of the coast of Britain. If you look at it from an airplane, its contours are not so different from the contours you see on the shore. The change in scaling does not alter the shapes or their degree of smoothness.

Pearls to Swine

What does fractal geometry have to do with the distribution of wealth, the size of cities, returns in the financial markets, the number of casualties in war, or the size of planets? Let us connect the dots.

The key here is that *the fractal has numerical or statistical measures that are (somewhat) preserved across scales*—the ratio is the same, unlike the Gaussian. Another view of such self-similarity is presented in [Figure 13](#). As we saw in [Chapter 15](#), the superrich are similar to the rich, only richer—wealth is scale independent, or, more precisely, of unknown scale dependence.

In the 1960s Mandelbrot presented his ideas on the prices of commodities and financial securities to the economics establishment, and the financial economists got all excited. In 1963 the then dean of the University of Chicago Graduate School of Business, George Shultz, offered him a professorship.